

GAIRA V6 — MSS Reference Manual

Seventeen molecular spectral signatures, and the evidence behind each

An MSS motif is a **validated spectral pattern**, not a molecule. Motif definitions (name, characteristic bands, exemplar chemistries) are curated from textbook Raman spectroscopy. Everything quantitative — which components express a motif, the weights, the confidence — is **derived** from the frozen atlas. In V6 no theme label enters any of it.

derivation parameter	value
weights	{'band': 0.3, 'basis_cosine': 0.3, 'exemplar': 0.3, 'perturbation': 0.1}
band_tolerance_cm	16.0
band_sigma_cm	9.0
keep_threshold	0.12
max_contributors	6
theme_evidence_used	False

The 17 V6 MSS motifs — implied Raman spectrum (red, $M[:,m]^T H$) against the declared band profile (blue dashed)

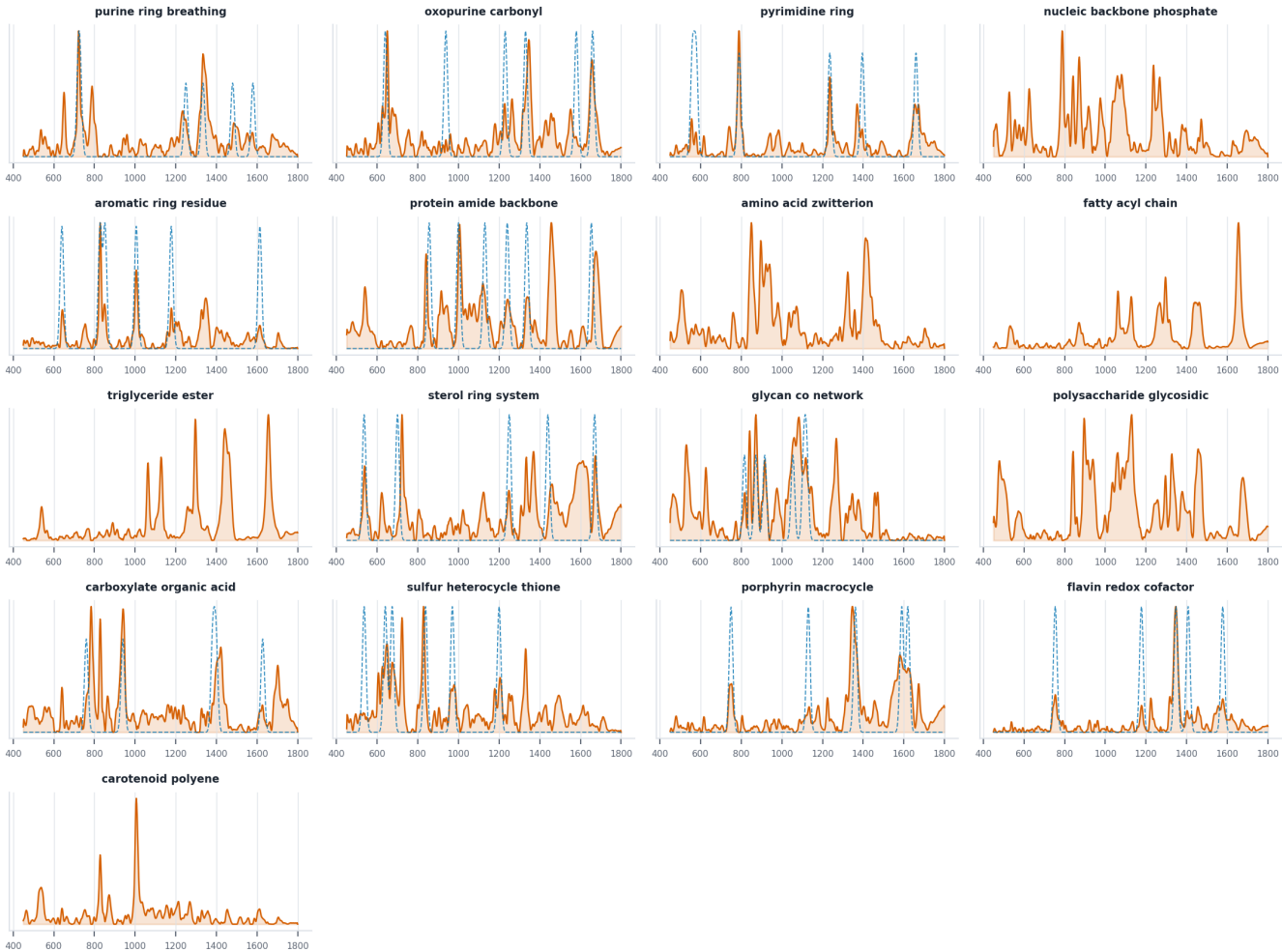


Figure 1 — Every motif's implied Raman spectrum (red) against its declared band profile (blue dashed).

Purine ring breathing

purine_ring_breathing · chemical class

Ring-breathing and stretching modes of the purine scaffold (adenine-type). The dominant SERS reporter for nucleic-purine chemistry; strong Ag adsorber.

Spectral justification

Characteristic bands (cm^{-1}): **720, 730, 1250, 1334, 1480, 1580**. Matching is region-based ($\pm 16 \text{ cm}^{-1}$), never exact-peak, and is additionally scored by the cosine between each component's frozen basis spectrum and a Gaussian profile built from these bands.

Supporting components

component	weight	band	basis cos	exemplar	pert	matched analytes
c3	0.355	0.83	0.805	0.91	1.00	adenine
c15	0.218	0.33	0.103	1.00	1.00	guanine, xanthine
c0	0.121	0.33	0.315	0.00	1.00	—
c2	0.110	0.33	0.225	0.00	1.00	—
c13	0.099	0.33	0.138	0.00	1.00	—
c18	0.097	0.33	0.124	0.00	1.00	—

Supporting analytes and coverage

Exemplar chemistries: adenine, guanine, hypoxanthine, xanthine

confidence	stability	breadth	purity	corpus analytes	discriminative AUC	band fidelity
0.683	0.819	0.833	0.077	4	0.983	0.701

Oxopurine carbonyl

oxopurine_carbonyl · chemical class

2,6-dioxopurine ring + carbonyl modes shared by xanthine/urate/guanine. The subgroup that responds when purines are oxidised (e.g. uricase depletion).

Spectral justification

Characteristic bands (cm^{-1}): **640, 938, 1230, 1330, 1580, 1660**. Matching is region-based ($\pm 16 \text{ cm}^{-1}$), never exact-peak, and is additionally scored by the cosine between each component's frozen basis spectrum and a Gaussian profile built from these bands.

Supporting components

component	weight	band	basis cos	exemplar	pert	matched analytes
c15	0.374	0.67	0.398	1.00	0.33	guanine, xanthine
c20	0.154	0.17	0.224	0.51	0.00	urate
c0	0.132	0.33	0.324	0.00	0.33	—
c7	0.124	0.33	0.385	0.00	0.00	—
c2	0.113	0.33	0.270	0.00	0.17	—

component	weight	band	basis cos	exemplar	pert	matched analytes
c17	0.102	0.17	0.233	0.19	0.00	urate

Supporting analytes and coverage

Exemplar chemistries: xanthine, guanine, urate, hypoxanthine

confidence	stability	breadth	purity	corpus analytes	discriminative AUC	band fidelity
0.623	0.831	0.750	0.072	4	0.883	0.617

Pyrimidine ring

pyrimidine_ring · chemical class

Ring modes of the pyrimidine scaffold (uracil / cytosine / thymine). Weaker Ag adsorber than purines; distinguished by the $\sim 790\text{ cm}^{-1}$ ring mode.

Spectral justification

Characteristic bands (cm^{-1}): **560, 578, 790, 1236, 1396, 1660**. Matching is region-based ($\pm 16\text{ cm}^{-1}$), never exact-peak, and is additionally scored by the cosine between each component's frozen basis spectrum and a Gaussian profile built from these bands.

Supporting components

component	weight	band	basis cos	exemplar	pert	matched analytes
c17	0.372	0.83	0.648	1.00	0.00	cytosine, uracil
c13	0.229	0.33	0.195	1.00	0.00	thymine
c23	0.182	0.33	0.366	0.52	0.00	cytosine
c7	0.078	0.17	0.355	0.00	0.00	—
c14	0.070	0.33	0.135	0.00	0.00	—
c21	0.068	0.33	0.122	0.00	0.00	—

Supporting analytes and coverage

Exemplar chemistries: uracil, cytosine, thymine, uridine

confidence	stability	breadth	purity	corpus analytes	discriminative AUC	band fidelity
0.487	0.780	0.625	0.116	3	1.000	0.693

Aromatic amino-acid residue

aromatic_ring_residue · chemical class

Aromatic side-chain ring modes — phenylalanine 1004 cm^{-1} ring breathing, tyrosine 828/850 Fermi doublet, tryptophan ring modes.

Spectral justification

Characteristic bands (cm^{-1}): **640, 828, 852, 1006, 1178, 1614**. Matching is region-based ($\pm 16 \text{ cm}^{-1}$), never exact-peak, and is additionally scored by the cosine between each component's frozen basis spectrum and a Gaussian profile built from these bands.

Supporting components

component	weight	band	basis cos	exemplar	pert	matched analytes
c5	0.331	0.67	0.625	1.00	0.00	phenylalanine, tyrosine
c6	0.212	0.17	0.417	0.89	0.00	phenylalanine, tryptophan
c11	0.169	0.17	0.406	0.60	0.00	alanine
c22	0.106	0.33	0.102	0.30	0.00	tryptophan
c15	0.092	0.17	0.247	0.22	0.00	tyrosine
c19	0.091	0.50	0.132	0.00	0.00	—

Supporting analytes and coverage

Exemplar chemistries: phenylalanine, tyrosine, tryptophan

confidence	stability	breadth	purity	corpus analytes	discriminative AUC	band fidelity
0.569	0.803	0.708	0.068	4	0.982	0.779

Protein amide backbone

protein_amide_backbone · chemical class

Peptide backbone signature — amide I (~ 1650), amide III (~ 1240), $\text{C}\alpha$ -C and backbone skeletal modes. Broad, mixture-dominated in real proteins.

Spectral justification

Characteristic bands (cm^{-1}): **856, 1000, 1130, 1240, 1336, 1654**. Matching is region-based ($\pm 16 \text{ cm}^{-1}$), never exact-peak, and is additionally scored by the cosine between each component's frozen basis spectrum and a Gaussian profile built from these bands.

Supporting components

component	weight	band	basis cos	exemplar	pert	matched analytes
c2	0.399	0.83	0.336	1.00	0.17	superoxide dismutases, trypsin, trypsinogen
c11	0.139	0.33	0.387	0.00	0.17	—
c0	0.127	0.33	0.317	0.00	0.17	—
c18	0.120	0.33	0.335	0.00	0.00	—
c7	0.109	0.17	0.438	0.00	0.00	—
c10	0.107	0.33	0.205	0.00	0.17	—

Supporting analytes and coverage

Exemplar chemistries: albumin, ubiquitin, trypsin, superoxide dismutases

confidence	stability	breadth	purity	corpus analytes	discriminative AUC	band fidelity
0.630	0.890	0.708	0.086	9	0.934	0.563

Lipid acyl chain

lipid_acyl_chain · chemical class

Aliphatic acyl-chain modes — CH₂ twist (~1300), scissor (~1440), C-C skeletal (~1060/1130), C=C (~1654). Shared by fatty acids and triglycerides.

Spectral justification

Characteristic bands (cm⁻¹): **1062, 1130, 1260, 1302, 1442, 1654**. Matching is region-based (± 16 cm⁻¹), never exact-peak, and is additionally scored by the cosine between each component's frozen basis spectrum and a Gaussian profile built from these bands.

Supporting components

component	weight	band	basis cos	exemplar	pert	matched analytes
c7	0.271	0.67	0.573	0.57	0.00	arachidonic acid
c16	0.222	0.50	0.626	0.35	0.00	palmitic acid
c1	0.191	0.33	0.501	0.44	0.00	elaidic acid, tripalmitolein
c10	0.113	0.50	0.255	0.00	0.00	—
c12	0.104	0.50	0.192	0.00	0.00	—
c2	0.098	0.33	0.320	0.00	0.00	—

Supporting analytes and coverage

Exemplar chemistries: elaidic acid, arachidonic acid, palmitic acid, tripalmitolein

confidence	stability	breadth	purity	corpus analytes	discriminative AUC	band fidelity
0.550	0.879	0.625	0.043	4	0.886	0.809

Sterol ring system

sterol_ring_system · chemical class

Fused steroid ring skeletal modes (cholesterol / estrone family). Weakly represented in the reference corpus — a low-coverage motif, flagged as such.

Spectral justification

Characteristic bands (cm⁻¹): **536, 700, 1250, 1440, 1670**. Matching is region-based (± 16 cm⁻¹), never exact-peak, and is additionally scored by the cosine between each component's frozen basis spectrum and a Gaussian profile built from these bands.

Supporting components

component	weight	band	basis cos	exemplar	pert	matched analytes
c3	0.244	0.40	0.185	0.61	0.00	estrone
c7	0.189	0.60	0.325	0.00	0.00	—
c2	0.182	0.40	0.491	0.00	0.00	—
c8	0.134	0.00	0.029	0.57	0.17	diethylstilbestrol
c15	0.126	0.40	0.161	0.00	0.17	—

component	weight	band	basis cos	exemplar	pert	matched analytes
c1	0.124	0.20	0.405	0.00	0.00	—

Supporting analytes and coverage

Exemplar chemistries: estrone, cholesterol, diethylstilbestrol

confidence	stability	breadth	purity	corpus analytes	discriminative AUC	band fidelity
0.524	0.899	0.583	0.018	3	0.683	0.491

Glycan C-O network

glycan_co_network · chemical class

Carbohydrate C-O / C-C / C-O-C ring and anomeric modes. Congested 450-1150 cm⁻¹ region shared across mono- and poly-saccharides.

Spectral justification

Characteristic bands (cm⁻¹): **816, 872, 916, 1054, 1108, 1122**. Matching is region-based (± 16 cm⁻¹), never exact-peak, and is additionally scored by the cosine between each component's frozen basis spectrum and a Gaussian profile built from these bands.

Supporting components

component	weight	band	basis cos	exemplar	pert	matched analytes
c10	0.201	0.67	0.425	1.00	0.00	d glucose, glucose
c18	0.190	0.50	0.484	1.00	0.00	cellulose, lactose
c4	0.188	0.50	0.463	1.00	0.00	fructose, galactose
c20	0.154	0.33	0.214	1.00	0.17	fructose
c12	0.150	0.33	0.227	1.00	0.00	d glucose, glucose
c21	0.118	0.33	0.110	0.78	0.00	fructose 6 phosphate

Supporting analytes and coverage

Exemplar chemistries: glucose, galactose, fructose, cellulose

confidence	stability	breadth	purity	corpus analytes	discriminative AUC	band fidelity
0.610	0.770	0.792	0.009	12	0.911	0.597

Carboxylate / organic acid

carboxylate_organic_acid · chemical class

Carboxylate symmetric stretch (~1400) and C-COO / skeletal modes of small TCA-cycle and metabolic acids.

Spectral justification

Characteristic bands (cm⁻¹): **760, 940, 1382, 1398, 1628**. Matching is region-based (± 16 cm⁻¹), never exact-peak, and is additionally scored by the cosine between each component's frozen basis spectrum and a Gaussian profile built from these bands.

Supporting components

component	weight	band	basis cos	exemplar	pert	matched analytes
c23	0.339	0.80	0.214	1.00	0.00	citrate, phosphoenolpyruvate, pyruvate
c14	0.334	0.60	0.468	0.91	0.00	citrate, succinic acid
c22	0.164	0.20	0.306	0.47	0.00	fumarate
c17	0.087	0.40	0.116	0.00	0.00	—
c13	0.077	0.20	0.255	0.00	0.00	—

Supporting analytes and coverage

Exemplar chemistries: citrate, succinic acid, pyruvate, fumarate

confidence	stability	breadth	purity	corpus analytes	discriminative AUC	band fidelity
0.496	0.764	0.650	0.106	5	0.972	0.513

Sulfur heterocycle / thione

sulfur_heterocycle_thione · chemical class

Thione / thiol and C-S modes — the ergothioneine thione ring and cysteine/ glutathione thiols. Strong Ag chemisorbers; the cleanest SERS antioxidant reporter.

Spectral justification

Characteristic bands (cm^{-1}): **536, 640, 674, 838, 970, 1200**. Matching is region-based ($\pm 16 \text{ cm}^{-1}$), never exact-peak, and is additionally scored by the cosine between each component's frozen basis spectrum and a Gaussian profile built from these bands.

Supporting components

component	weight	band	basis cos	exemplar	pert	matched analytes
c19	0.263	0.67	0.477	0.53	0.32	cysteine, ergothioneine
c15	0.191	0.17	0.324	0.66	0.44	cysteine, methionine
c21	0.171	0.33	0.383	0.44	0.00	ergothioneine
c5	0.137	0.33	0.432	0.00	0.48	—
c3	0.120	0.17	0.113	0.43	0.31	methionine
c20	0.120	0.17	0.229	0.32	0.30	cysteine

Supporting analytes and coverage

Exemplar chemistries: ergothioneine, cysteine, glutathione, methionine

confidence	stability	breadth	purity	corpus analytes	discriminative AUC	band fidelity
0.723	0.789	0.917	0.024	5	0.804	0.639

Porphyrin macrocycle

porphyrin_macrocycle · chemical class

Resonance-enhanced tetrapyrrole macrocycle modes of heme proteins (hemoglobin, myoglobin, cytochrome c) — pyrrole breathing and $\text{C}\alpha\text{-Cm}$ stretches.

Spectral justification

Characteristic bands (cm^{-1}): **750, 1130, 1362, 1590, 1620**. Matching is region-based ($\pm 16 \text{ cm}^{-1}$), never exact-peak, and is additionally scored by the cosine between each component's frozen basis spectrum and a Gaussian profile built from these bands.

Supporting components

component	weight	band	basis cos	example	pert	matched analytes
c8	0.328	0.60	0.655	1.00	0.00	cytochrome c, hemoglobin, melanin
c13	0.238	0.40	0.394	0.84	0.00	cytochrome c, hemoglobin, myoglobin
c0	0.212	0.60	0.352	0.50	0.00	hemoglobin, myoglobin
c22	0.077	0.40	0.129	0.00	0.00	—
c18	0.073	0.20	0.303	0.00	0.00	—
c2	0.073	0.40	0.100	0.00	0.00	—

Supporting analytes and coverage

Exemplar chemistries: hemoglobin, myoglobin, cytochrome c, melanin

confidence	stability	breadth	purity	corpus analytes	discriminative AUC	band fidelity
0.515	0.825	0.625	0.099	4	0.995	0.681

Flavin / redox cofactor

flavin_redox_cofactor · chemical class

Isoalloxazine (flavin) and related redox-cofactor ring modes — riboflavin/ FAD, ubiquinone. Resonance-enhanced; the dominant redox reporter.

Spectral justification

Characteristic bands (cm^{-1}): **754, 1178, 1348, 1408, 1578**. Matching is region-based ($\pm 16 \text{ cm}^{-1}$), never exact-peak, and is additionally scored by the cosine between each component's frozen basis spectrum and a Gaussian profile built from these bands.

Supporting components

component	weight	band	basis cos	example	pert	matched analytes
c0	0.432	0.80	0.645	1.00	0.00	riboflavin
c22	0.183	0.60	0.438	0.00	0.00	—
c8	0.123	0.40	0.298	0.00	0.00	—
c17	0.109	0.20	0.057	0.36	0.00	riboflavin
c2	0.082	0.40	0.065	0.00	0.00	—
c13	0.071	0.20	0.202	0.00	0.00	—

Supporting analytes and coverage

Exemplar chemistries: riboflavin, ubiquinone, nadh, fad

confidence	stability	breadth	purity	corpus analytes	discriminative AUC	band fidelity
0.475	0.814	0.583	0.126	2	1.000	0.692

Colloid / matrix background

colloid_matrix_background · chemical class · non-biochemical

Non-biochemical substrate + matrix contribution (citrate capping, buffer, serum background).
Reported for honesty; NOT a biochemical motif.

Spectral justification

Characteristic bands (cm^{-1}): **930, 1050, 1400**. Matching is region-based ($\pm 16 \text{ cm}^{-1}$), never exact-peak, and is additionally scored by the cosine between each component's frozen basis spectrum and a Gaussian profile built from these bands.

Supporting components

component	weight	band	basis cos	exemplar	pert	matched analytes
c14	0.279	0.67	0.543	0.97	0.00	citrate, phosphate
c21	0.180	0.33	0.080	1.00	0.00	fructose 6 phosphate, ure
c23	0.177	0.33	0.116	0.88	0.17	citrate
c6	0.131	0.00	0.022	1.00	0.00	ure, urea
c10	0.125	0.67	0.315	0.00	0.00	—
c9	0.108	0.67	0.176	0.00	0.00	—

Supporting analytes and coverage

Exemplar chemistries: citrate, phosphate, urea

confidence	stability	breadth	purity	corpus analytes	discriminative AUC	band fidelity
0.477	0.763	0.625	0.030	6	0.941	0.451

Known limitations of the motif set

- **sterol_ring_system** was rebuilt on canonical cholesterol bands but its discriminative AUC is still the lowest of the set (0.66) and its top activators are not sterols. No component in the frozen atlas isolates the steroid ring system — this is a foundation limit, not a definition problem.
- **porphyrin_macrocycle** and **flavin_redox_cofactor** are marked low_coverage: the corpus contains no pure porphyrin or flavin reference, so both borrow protein and purine components. Their component-support cosine is 0.70.
- **carotenoid_polyene** rests on two corpus analytes.
- **protein_amide_backbone** has 32 supporting analytes but its bands overlap the saccharide C-O region; its top activators are sugars.
- **Spectral purity is low across the board** (0.01–0.13): every motif is supported diffusely across its (up to six) components rather than concentrated on one. That is a property of a 24-component basis fitted to 167 analytes, and it is why motifs — not components — are the right unit of interpretation.

Atlas 09ed804a40836f4a05a91ba10900cded. Spec: results/v6_rebuild/artifacts/mss_motifs_v6.yaml